

# When Do Physics Priors Help?

## A Controlled Benchmark of Physics-Informed Dynamics Models for Underactuated Swing-Up

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**Abstract**—Physics-informed machine learning promises data-efficient, physically consistent dynamics models, but it is often unclear *when* a physics prior pays off relative to a black-box baseline—or whether a mismatched prior can hurt. We present a controlled, reproducible benchmark of seven learned dynamics models—a multilayer perceptron (MLP), a polynomial system-identification baseline, a Gaussian process (GP), a Lagrangian Neural Network (LNN), a naive Hamiltonian Neural Network (HNN), a structured HNN with a learned mass matrix, and a Lagrangian-informed GP (LGP)—on two underactuated systems, the Acrobot and the Pendubot. We evaluate four axes: one-step accuracy, long-horizon rollout stability (valid prediction time), energy drift, and downstream energy-based swing-up control, across five seeds, three data budgets, and both clean and sensor/process-noise conditions. We find a nuanced map rather than a single winner. The LNN, whose learned positive-definite mass matrix matches these systems, is the most accurate and energy-consistent and never diverges. The GP is the most data-efficient and yields the most stable rollouts despite modest one-step accuracy. The LGP blends the two, achieving LNN-level accuracy with GP-level rollout stability and the lowest energy drift. The naive HNN, whose identity-mass assumption is violated by configuration-dependent inertia, is worse than the black-box MLP and diverges in 20–32% of rollouts; a controlled ablation shows that giving the Hamiltonian model a learned mass matrix—which by Legendre duality recovers the LNN—fully removes this failure, isolating the assumption rather than the formulation as the cause. Under noise all models degrade gracefully, but the GP residual in the LGP overfits noise and loses rollout stability. Only models with an energy function can drive energy-based control. The result is a concrete map of when physics priors help: scarce data, long horizons and energy-sensitive tasks, and a prior whose structure matches the system.

**Index Terms**—physics-informed machine learning, learned dynamics, underactuated robots, Lagrangian neural networks, Gaussian processes, swing-up control.

### I. INTRODUCTION

Learning the dynamics of a mechanical system from data underpins model-based control and reinforcement learning. Plain neural networks fit one-step transitions accurately, yet rolled out over long horizons they drift and violate invariants such as energy conservation. Physics-informed machine learning (PiML) bakes structure—Lagrangian or Hamiltonian

mechanics, positive-definite inertia, energy conservation—into the model [1], [2], promising better long-horizon behaviour and sample efficiency.

In practice it is rarely obvious *when* a prior is worth its complexity, or whether a prior whose assumptions are violated by the real system can hurt. Most papers introduce one physics-informed model and show it beats a baseline on a favourable task; what is missing is a controlled, like-for-like comparison across priors, evaluation criteria, data budgets, and noise conditions. Underactuated systems such as the Acrobot and Pendubot are an ideal stress test: they are low-dimensional yet chaotic, have configuration-dependent inertia, and admit a demanding downstream control task (swing-up) that depends on the learned model’s physical consistency, not just its one-step error.

We provide such a comparison and summarise it as a map of when physics priors help, with four findings:

- 1) **Matching structure matters most.** The LNN, which learns a positive-definite, configuration-dependent mass matrix, is the most accurate and energy-consistent model and never diverges.
- 2) **A mismatched prior is worse than none.** The naive HNN, assuming an identity mass matrix, is worse than the black-box MLP and diverges in 20–32% of rollouts. A controlled ablation pinpoints the assumption: giving the Hamiltonian model a learned mass matrix removes the failure.
- 3) **Uncertainty buys stability and data efficiency.** The GP is the most data-efficient model and yields the most stable rollouts; combining it with Lagrangian structure (the LGP) gives the best overall blend on clean data.
- 4) **Robustness has costs.** Under noise the GP is the most robust, while the LGP’s GP residual overfits noise and loses rollout stability.

Our contribution is the controlled study and the resulting map, together with a released, configuration-driven pipeline that regenerates every result.

## II. RELATED WORK

**Physics-informed neural dynamics.** Deep Lagrangian Networks (DeLaN) [3] learn the mass matrix and potential of a Lagrangian and recover the equations of motion by automatic differentiation; Lagrangian Neural Networks [4] generalise the idea to an arbitrary learned Lagrangian. Hamiltonian Neural Networks [5] learn a Hamiltonian and integrate the symplectic flow, conserving energy by construction but requiring canonical coordinates; symplectic and variational-integrator variants extend them to control and rigid bodies [6], [7]. Physics-informed neural networks more broadly embed governing equations into the loss [2], [1].

**Gaussian-process and classical dynamics.** GPs provide calibrated uncertainty and data efficiency for model learning [8], [9] and underpin sample-efficient control such as PILCO [10]. Sparse regression on a feature library (SINDy) is a classical system-identification baseline [11]. Hybrid schemes learn a data-driven residual on top of a parametric or structured model; our Lagrangian-informed GP is an instance with a learned-Lagrangian mean.

**Underactuated swing-up.** The Acrobot and Pendubot are standard benchmarks; energy-shaping controllers swing them toward upright by regulating total energy, typically with an LQR catch for final balancing [12], [13]. We use swing-up as a downstream task that stresses the *usefulness* of a learned model, not only its predictive error.

Relative to this literature we hold data, integrator, training objective, and evaluation fixed across seven models and report multiple axes, data budgets and noise conditions, enabling a fair comparison rather than a single favourable demonstration.

## III. BACKGROUND AND PROBLEM SETUP

### A. Underactuated double pendulums

We study two planar double pendulums with a single actuator. Their dynamics follow the manipulator equation

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = Bu, \quad (1)$$

with joint angles  $q = (q_1, q_2)$ , symmetric positive-definite mass matrix  $M(q) \succ 0$ , Coriolis/centrifugal term  $C(q, \dot{q})$ , gravity  $g(q) = \partial V / \partial q$  derived from a potential  $V(q)$ , and actuation matrix  $B \in \mathbb{R}^2$ . The **Acrobot** is actuated only at the elbow ( $B = [0, 1]^\top$ ) and the **Pendubot** only at the shoulder ( $B = [1, 0]^\top$ ); the remaining joint is passive, making both systems underactuated. The state is  $x = (q_1, q_2, \dot{q}_1, \dot{q}_2)$  and the control  $u$  is a scalar torque. The total mechanical energy is  $E(x) = \frac{1}{2}\dot{q}^\top M(q)\dot{q} + V(q)$ , which is conserved when  $u = 0$ . We integrate the true dynamics with a fourth-order Runge–Kutta (RK4) scheme at  $\Delta t = 10^{-2}$  s.

### B. Lagrangian and Hamiltonian structure

The same dynamics admit two structured parameterizations that the models below exploit. The Lagrangian  $\mathcal{L} = \frac{1}{2}\dot{q}^\top M(q)\dot{q} - V(q)$  yields the Euler–Lagrange equations, equivalent to (1) with  $C\dot{q} = \dot{M}\dot{q} - \frac{1}{2}\nabla_q(\dot{q}^\top M\dot{q})$ . The Hamiltonian  $H(q, p) = \frac{1}{2}p^\top M(q)^{-1}p + V(q)$ , with canonical momentum  $p = M(q)\dot{q}$ , yields Hamilton’s equations

$\dot{q} = \partial H / \partial p$ ,  $\dot{p} = -\partial H / \partial q + Bu$ . The two are related by the Legendre transform, so a model that learns  $M(q)$  and  $V(q)$  produces the *same* equations of motion whether written in Lagrangian or Hamiltonian form—a fact we use to construct a fair Hamiltonian baseline.

### C. Learning problem

Given transitions  $(x_t, u_t, x_{t+1})$  collected under random torques, the task is to learn  $\dot{f}$  predicting the state derivative (hence  $x_{t+1}$  via the same RK4 integrator). A single integrator and one-step objective are used for every model, so behavioural differences are attributable to the prior, not the training setup.

## IV. THE MODEL ZOO

All models map  $(x, u)$  to a state derivative and are rolled out with RK4. Neural models use two hidden layers of 64 units; Lagrangian/Hamiltonian models use Softplus activations (twice differentiable, required for the autograd-derived forces).

**Vanilla MLP (black box).** A 128-128-128 network regresses joint accelerations directly; no physical structure.

**Polynomial (system ID).** Closed-form ridge regression of accelerations on a degree-2 polynomial expansion of the features  $\phi = [\sin q_1, \cos q_1, \sin q_2, \cos q_2, \dot{q}_1, \dot{q}_2, u]$ ; a cheap classical baseline ( $\ell_2$  penalty  $10^{-4}$ ).

**Gaussian Process (GP).** An exact GP with an ARD-RBF kernel regresses each acceleration component from  $\phi$ , fit by marginal likelihood; the  $(\sin, \cos)$  encoding respects angle periodicity. We cap the training set at 1000 points for tractability of the  $O(n^3)$  solve.

**Lagrangian Neural Network (LNN).** Two networks output a Cholesky factor of  $M(q) = L(q)L(q)^\top + \epsilon I \succ 0$  and a potential  $V(q)$ ; the Coriolis and gravity terms follow by automatic differentiation and the acceleration from (1). The known  $B$  is supplied as a prior.

**Naive HNN.** A scalar Hamiltonian  $H(q, p)$  with the identity-mass convention  $p := \dot{q}$ , integrated along the symplectic flow. It does *not* learn a configuration-dependent mass matrix.

**Structured HNN.** A Hamiltonian model that *learns*  $M(q)$  and  $V(q)$  with canonical momentum  $p = M(q)\dot{q}$ . By the Legendre transform its equations of motion coincide with the LNN’s; we include it as a controlled ablation that isolates the naive HNN’s identity-mass assumption.

**Lagrangian-informed GP (LGP).** A hybrid: an LNN supplies a physics mean and a GP models the residual acceleration,  $\ddot{q} = \ddot{q}_{\text{LNN}} + \text{GP}_{\text{res}}$ , keeping structure while letting the GP correct what the structural model misses.

### A. Structured dynamics in detail

The LNN and Structured HNN share the same learned ingredients. To guarantee a physically valid, invertible inertia, the mass matrix is parameterized through a Cholesky factor,

$$M(q) = L(q)L(q)^\top + \epsilon I, \quad (2)$$

where  $L(q)$  is lower-triangular with positive (Softplus) diagonal and  $\epsilon = 10^{-2}$  floors the smallest eigenvalue for a well-conditioned solve. Given  $M(q)$  and  $V(q)$ , the gravity and Coriolis terms are obtained by automatic differentiation,

$$g(q) = \nabla_q V, \quad [C(q, \dot{q})\dot{q}]_i = (\dot{M}\dot{q})_i - \frac{1}{2} \dot{q}^\top \frac{\partial M}{\partial q_i} \dot{q}, \quad (3)$$

with  $\dot{M} = \sum_k (\partial M / \partial q_k) \dot{q}_k$ , and the acceleration solves  $M\ddot{q} = Bu - C\dot{q} - g$ . The Structured HNN evaluates the algebraically equivalent Hamiltonian form  $\ddot{q} = M^{-1}(-\partial H / \partial q|_p + Bu - \dot{M}\dot{q})$ ; both conserve their own learned energy to  $\sim 10^{-7}$  over 200 steps even before training, confirming the correctness of the autograd equations of motion. The naive HNN omits  $M(q)$ , setting  $p := \dot{q}$ .

## V. EVALUATION PROTOCOL

**Data.** Per system we generate 8000 transitions from random short rollouts under uniform random torques; a 20% tail is held out for one-step evaluation and the remaining pool is randomly subsampled to budgets of 250, 1000, 4000 to probe data efficiency. The *noisy* condition adds process noise—a Gaussian state kick after each step, modelling unmodelled forces—and sensor noise added to the recorded states, both with standard deviation 0.02.

**Training.** Each model minimises a one-step angle-aware MSE between the RK4-integrated prediction and the recorded next state (GP-style models are instead fit by marginal likelihood). The angle-aware loss wraps angular errors to  $[-\pi, \pi]$  so the  $\pm\pi$  discontinuity does not corrupt the gradient. We use 100 epochs, Adam, gradient clipping, and *five* seeds per configuration.

**Implementation.** The MLP has three 128-unit ReLU layers; the structured networks use two 64-unit Softplus layers each for  $M(q)$  and  $V(q)$ . We train with Adam (learning rate  $10^{-3}$ , batch size 256) and clip gradient norms to 10. The GP uses an ARD-RBF kernel with per-feature lengthscales and signal/noise variances optimised by 25 marginal-likelihood steps in double precision; the LGP fits its residual GP on the same capped subsample. Process and sensor noise are zero-mean Gaussian (std 0.02) on the four state components, and a rollout is declared diverged if any state exceeds  $10^2$  in magnitude or becomes non-finite. All experiments run on CPU.

**Metrics.** (1) *One-step MSE* on the held-out set. (2) *Valid prediction time*: the number of steps an autoregressive free-swing ( $u=0$ ) rollout tracks the truth before the per-step error first exceeds a fixed threshold. On chaotic systems raw long-horizon MSE saturates (two nearby trajectories decorrelate and the error reaches a ceiling), so it cannot discriminate models; valid prediction time does. (3) *Energy drift*: the range of the true energy  $E(x)$  along the model’s free rollout, which should be zero; we also report the fraction of rollouts that diverge (state magnitude or non-finite values exceeding a bound). (4) *Swing-up control* (Section VI-F). All rollout evaluations use identical fixed initial conditions across models for fairness.

TABLE I  
ONE-STEP MSE VS. DATA BUDGET (ACROBOT, CLEAN, MEAN OVER FIVE SEEDS). PHYSICS/GP MODELS DOMINATE AT LOW DATA; THE MLP CATCHES UP ONLY WITH MORE DATA.

| Model         | $n=250$                                | $n=1000$                               | $n=4000$                               |
|---------------|----------------------------------------|----------------------------------------|----------------------------------------|
| Vanilla MLP   | $7.3 \times 10^{-2}$                   | $4.7 \times 10^{-2}$                   | $6.1 \times 10^{-3}$                   |
| Polynomial    | $5.1 \times 10^{-2}$                   | $4.4 \times 10^{-2}$                   | $4.2 \times 10^{-2}$                   |
| Vanilla GP    | $2.1 \times 10^{-2}$                   | $8.4 \times 10^{-3}$                   | $8.5 \times 10^{-3}$                   |
| Naive HNN     | $8.3 \times 10^{-2}$                   | $4.6 \times 10^{-2}$                   | $1.6 \times 10^{-2}$                   |
| Lagrangian NN | $2.7 \times 10^{-2}$                   | $8.5 \times 10^{-3}$                   | <b><math>1.4 \times 10^{-3}</math></b> |
| Lagrangian GP | <b><math>1.9 \times 10^{-2}</math></b> | <b><math>2.9 \times 10^{-3}</math></b> | $1.5 \times 10^{-3}$                   |

## VI. RESULTS

### A. One-step accuracy and data efficiency

Table I reports one-step MSE versus data budget on the Acrobot. At the smallest budget (250 points) the GP, LGP and LNN are already an order of magnitude more accurate than the MLP and naive HNN; the MLP only becomes competitive at 4000 points. The Polynomial baseline barely improves with data (its fixed feature basis underfits), and the GP plateaus once it hits its 1000-point cap. Physics structure and GP smoothing thus pay off most precisely in the low-data regime—the regime that matters for real robots.

### B. Long-horizon stability and energy

Table II reports all three dynamics axes at the largest budget, and Fig. 1 the full sweep. Fig. 2 illustrates the mechanism: on a free swing both the MLP and LNN trajectories eventually decorrelate from the truth (chaos), but the MLP’s *energy* balloons while the LNN’s stays near the conserved value—the LNN’s error is physically constrained.

The LNN is the most accurate model and among the lowest in energy drift on both systems. On the chaotic free-swing task the GP gives by far the longest valid prediction time ( $> 2.5 \times$  the MLP) despite larger one-step error: smoothing toward the GP prior damps the error growth that destabilises the MLP. The LGP combines LNN-level accuracy with high valid time and the lowest energy drift, making it the best all-round model on clean dynamics; on the Pendubot it more than doubles the LNN’s valid time. The Polynomial baseline is weakest, with large drift and 16–36% divergence.

### C. The HNN failure is the assumption, not the formulation

The naive HNN is the worst structured model: poor one-step accuracy, the fewest valid steps, and 20–32% divergence (its huge drift values are an artifact of these divergences). Table III isolates the cause. Replacing the identity-mass assumption with a learned mass matrix—the Structured HNN, which by Legendre duality has the same equations of motion as the LNN—removes the failure entirely: divergence drops to 0% and valid steps and drift match the LNN. The naive HNN’s failure is therefore due to the violated identity-mass assumption, not to the Hamiltonian formulation. This is the practical lesson of the paper in miniature: a structured prior helps only when its structural assumptions hold.

TABLE II

DYNAMICS AXES (CLEAN,  $n=4000$ , MEAN OVER FIVE SEEDS). ARROWS MARK LOWER ( $\downarrow$ ) / HIGHER ( $\uparrow$ ) IS BETTER. “DIV.” IS THE FRACTION OF ROLLOUTS THAT DIVERGED. THE STRUCTURED HNN COINCIDES WITH THE LNN (LEGENDRE DUALITY) AND IS ANALYSED SEPARATELY IN TABLE III.

| System   | Model         | One-step MSE $\downarrow$              | Valid steps $\uparrow$ | Energy drift $\downarrow$ | Div. |
|----------|---------------|----------------------------------------|------------------------|---------------------------|------|
| Acrobot  | Vanilla MLP   | $6.1 \times 10^{-3}$                   | 25.8                   | $1.98 \times 10^1$        | 0%   |
|          | Polynomial    | $4.2 \times 10^{-2}$                   | 13.2                   | $6.7 \times 10^2$         | 36%  |
|          | Vanilla GP    | $8.5 \times 10^{-3}$                   | <b>67.6</b>            | 9.7                       | 0%   |
|          | Lagrangian NN | <b><math>1.4 \times 10^{-3}</math></b> | 27.3                   | 7.7                       | 0%   |
|          | Naive HNN     | $1.6 \times 10^{-2}$                   | 10.7                   | $9.6 \times 10^2$         | 20%  |
|          | Lagrangian GP | $1.5 \times 10^{-3}$                   | 52.9                   | <b>5.7</b>                | 0%   |
| Pendubot | Vanilla MLP   | $7.2 \times 10^{-3}$                   | 28.0                   | $1.87 \times 10^1$        | 0%   |
|          | Polynomial    | $4.1 \times 10^{-2}$                   | 11.9                   | $3.0 \times 10^2$         | 16%  |
|          | Vanilla GP    | $8.8 \times 10^{-3}$                   | <b>71.4</b>            | $1.13 \times 10^1$        | 0%   |
|          | Lagrangian NN | <b><math>2.6 \times 10^{-3}</math></b> | 26.9                   | 9.9                       | 0%   |
|          | Naive HNN     | $1.6 \times 10^{-2}$                   | 12.2                   | $1.0 \times 10^3$         | 32%  |
|          | Lagrangian GP | $2.3 \times 10^{-3}$                   | 58.6                   | <b>6.7</b>                | 0%   |

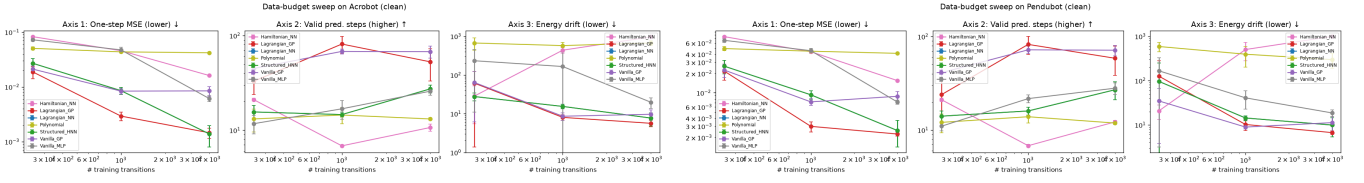


Fig. 1. Data-budget sweep (clean), Acrobot (left) and Pendubot (right): each dynamics axis versus the number of training transitions (log-log), mean over five seeds with standard-deviation error bars.

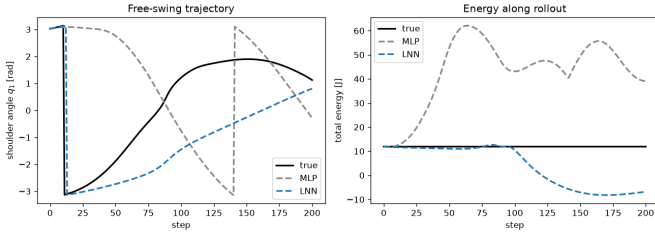


Fig. 2. Free-swing rollout on the Acrobot. Both learned trajectories eventually decorrelate from the truth (left, chaos), but the MLP’s total energy diverges while the LNN stays near the conserved value (right). Angles are wrapped to  $[-\pi, \pi]$ .

TABLE III

HNN ABLATION ON THE ACROBOT ( $n=4000$ , FIVE SEEDS). LEARNING THE MASS MATRIX REMOVES THE NAIVE HNN’S FAILURE; THE STRUCTURED HNN MATCHES THE LNN.

| Model                         | Valid steps $\uparrow$ | Energy drift $\downarrow$ | Div. |
|-------------------------------|------------------------|---------------------------|------|
| Naive HNN (identity mass)     | 10.7                   | $9.6 \times 10^2$         | 20%  |
| Structured HNN (learned mass) | 27.3                   | 7.7                       | 0%   |
| Lagrangian NN (reference)     | 27.3                   | 7.7                       | 0%   |

#### D. Robustness to noise

Table IV reports the clean  $\rightarrow$  noisy change for both systems at  $n=4000$ . All models degrade gracefully—one-step error roughly doubles—but two effects stand out. The GP remains the most stable rollout model in absolute terms. The LGP’s GP residual, by contrast, *overfits* the noise: its valid prediction time collapses (Acrobot  $52.9 \rightarrow 36.4$ , Pendubot  $58.6 \rightarrow 32.7$ ),

TABLE IV  
NOISE ROBUSTNESS AT  $n=4000$ : ONE-STEP MSE AND VALID STEPS, CLEAN  $\rightarrow$  NOISY (MEAN OVER FIVE SEEDS).

| System   | Model         | One-step MSE                            | Valid steps             |
|----------|---------------|-----------------------------------------|-------------------------|
| Acrobot  | Vanilla MLP   | $6.1 \rightarrow 10.2 (\times 10^{-3})$ | $25.8 \rightarrow 29.6$ |
|          | Vanilla GP    | $8.5 \rightarrow 13.4 (\times 10^{-3})$ | $67.6 \rightarrow 59.3$ |
|          | Lagrangian NN | $1.4 \rightarrow 3.3 (\times 10^{-3})$  | $27.3 \rightarrow 35.4$ |
|          | Lagrangian GP | $1.5 \rightarrow 3.3 (\times 10^{-3})$  | $52.9 \rightarrow 36.4$ |
| Pendubot | Vanilla MLP   | $7.2 \rightarrow 9.6 (\times 10^{-3})$  | $28.0 \rightarrow 28.3$ |
|          | Vanilla GP    | $8.8 \rightarrow 10.0 (\times 10^{-3})$ | $71.4 \rightarrow 60.0$ |
|          | Lagrangian NN | $2.6 \rightarrow 4.1 (\times 10^{-3})$  | $26.9 \rightarrow 28.4$ |
|          | Lagrangian GP | $2.3 \rightarrow 3.9 (\times 10^{-3})$  | $58.6 \rightarrow 32.7$ |

erasing the clean-condition advantage that made it the best all-rounder. Hard structural priors (LNN) are comparatively noise-tolerant, even gaining valid steps as the injected noise acts as mild regularization. The practical implication is that a learned residual is a liability under observation noise, whereas a pure structural prior degrades benignly.

#### E. Computational cost

Table V lists wall-clock training time (Acrobot,  $n=1000$ , 100 epochs, CPU). The closed-form Polynomial is essentially free; the MLP, GP and naive HNN are cheap. The structured models (LNN, Structured HNN, LGP) cost 3–4 $\times$  the MLP because each forward pass differentiates the learned mass matrix and potential. The LGP additionally trains an internal LNN, making it the most expensive. These costs are modest in absolute terms but should be weighed against the accuracy/stability gains when choosing a model.

TABLE V  
TRAINING TIME (ACROBOT,  $n=1000$ , 100 EPOCHS, SINGLE CPU).

| Model          | Training time (s) |
|----------------|-------------------|
| Polynomial     | 0.1               |
| Naive HNN      | 4.6               |
| Vanilla MLP    | 6.4               |
| Vanilla GP     | 6.9               |
| Structured HNN | 19.8              |
| Lagrangian NN  | 20.3              |
| Lagrangian GP  | 23.8              |

TABLE VI  
ENERGY-BASED SWING-UP (FIVE SEEDS). MEAN TIP HEIGHT AND SUCCESS RATE; THE ORACLE USES THE TRUE ENERGY. BLACK-BOX MODELS HAVE NO ENERGY FUNCTION (N/A). THE STRUCTURED HNN MATCHES THE LNN.

| System   | Model         | Mean tip height | Success    |
|----------|---------------|-----------------|------------|
| Acrobot  | Oracle        | 0.97            | 100%       |
|          | Lagrangian NN | <b>0.63</b>     | <b>25%</b> |
|          | Lagrangian GP | 0.63            | 25%        |
|          | Naive HNN     | 0.51            | 0%         |
| Pendubot | Oracle        | 0.96            | 100%       |
|          | Lagrangian NN | <b>0.65</b>     | <b>15%</b> |
|          | Lagrangian GP | 0.65            | 15%        |
|          | Naive HNN     | 0.66            | 10%        |

#### F. Downstream swing-up control

We drive an energy-shaping controller on the *true* plant using a model’s energy estimate,

$$u = k_e (E_{\text{top}} - E_{\text{model}}(x)) \dot{q}_{\text{act}} - k_p q_{\text{act}} - k_d \dot{q}_{\text{act}}, \quad (4)$$

clipped to a torque limit, where  $E_{\text{top}}$  is the upright energy under the *same* model (so any constant energy offset cancels) and  $q_{\text{act}}$  is the actuated joint. The law pumps energy toward the upright level while damping the actuated joint. Only models with an energy function (LNN, Structured HNN, Naive HNN, LGP) can drive it; the MLP, Polynomial and GP cannot, which is itself a result. We report the mean normalised tip height reached (0 down, 1 upright)—a continuous, robust measure of swing-up quality—and a success rate, with a true-energy Oracle as the ceiling (Table VI, Fig. 3).

Models with an accurate energy (LNN, LGP, Structured HNN) swing the tip to  $\sim 65\%$  of full height, well above the naive HNN, whose unreliable energy yields the worst control; the black-box models cannot attempt the task at all. No learned model matches the Oracle, and robustly *balancing* the underactuated systems at the top—which requires an LQR catch fed by a precise, low-velocity swing-up entry—remains hard; we therefore treat the continuous tip-height as the primary control metric and provide an optional LQR catch in the released code. The ranking nonetheless tracks energy-model quality, directly connecting predictive consistency to downstream usefulness.

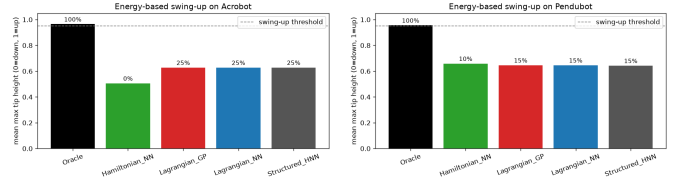


Fig. 3. Swing-up: bars are mean tip height, percentages are success rate. Black-box models are absent (no energy function).

## VII. DISCUSSION: A MAP OF WHEN PHYSICS PRIORS HELP

Across the four axes a consistent picture emerges. *First*, physics priors help most when their structure matches the system: the LNN’s learned positive-definite, configuration-dependent mass matrix is the single most valuable ingredient, driving its wins on accuracy, energy and control, and the Structured-HNN ablation shows the same structure rescues a failing model. *Second*, a mismatched prior is actively harmful—the naive HNN’s identity-mass assumption makes it worse than a black box and unstable. *Third*, the benefit of any prior is largest with scarce data and on long-horizon, energy-sensitive tasks; at high data with only one-step accuracy in view, the black-box MLP is competitive, at a fraction of the training cost. *Fourth*, uncertainty-aware models (GPs) buy rollout stability and data efficiency complementary to hard priors, and combining the two (the LGP) is best on clean dynamics—but its residual overfits noise, a robustness cost a practitioner must weigh. No single model wins everything; the practical takeaway is to match the prior to the data regime, horizon, noise level, and downstream task.

## VIII. LIMITATIONS AND FUTURE WORK

Our study is in simulation on two planar systems with a fixed noise level. Absolute swing-up success is modest because robustly balancing underactuated systems is a hard control problem in its own right; a model-predictive controller would let black-box models also be evaluated downstream and would better exploit accurate one-step predictions. The exact GP is capped in training-set size and is near-deterministic at the largest budget. Natural extensions include additional systems (e.g. cart-pole), real-robot data, a sweep over noise levels, sparse or deep-kernel GPs to scale the GP and LGP, and propagating GP/ensemble uncertainty into the controller.

## IX. CONCLUSION

We presented a controlled, reproducible benchmark of seven dynamics models on two underactuated systems across four axes, five seeds, three data budgets, and clean and noisy conditions. Rather than crowning a single method, our results map out *when* physics priors help: most when the prior matches the system’s structure, when data are scarce, and on long-horizon, energy-sensitive tasks—while a mismatched prior can be worse than none, and a learned residual can be a liability under noise. We release the pipeline as a reusable testbed for physics-informed dynamics learning.

## REPRODUCIBILITY

Code, configurations, datasets, and the scripts that regenerate every table and figure are available at <https://github.com/amirhossein-fattahi/piml-underactuated>.

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